1. Project Overview

1.1 Project Name

SRP8-043 – Front Suspension

1.2 Lead

John Zhu

1.3 Scope

This project encompasses the development of the front suspension geometry as well as the spring and damper system on SR8. This development includes optimising the subsystem's performance and the SR8 project's vision while complying with all necessary regulations.

1.4 Out of Scope

1.5 Assemblies

- ASSY - FRONT LOWER WISHBONE - LH
- ASSY - FRONT LOWER WISHBONE - RH
- ASSY - FRONT TIEROD - LH
- ASSY - FRONT TIEROD - RH
- ASSY - FRONT TOP WISHBONE - LH
- ASSY - FRONT TOP WISHBONE - RH

2. Interfacing Requirements (PMS)

2.1 FRONT LOWER WISHBONE

2.1 Dependencies (what this project relies on)

Assembly	Requirement	Type

2.2 Impacts (what relies on this project)

Assembly	Requirement	Type

2.2 FRONT TIEROD

2.1 Dependencies (what this project relies on)

Assembly	Requirement	Type

2.2 Impacts (what relies on this project)

Assembly	Requirement	Type

2.3 FRONT TOP WISHBONE

2.1 Dependencies (what this project relies on)

Assembly	Requirement	Type

2.2 Impacts (what relies on this project)

Assembly	Requirement	Type

3. ADR Requirements

I'm unsure if there may be other requirements relating to suspension, but the ones assigned apparently apply to trucks, trailers and multi-axle vehicles rather than SR8 which has independent suspension (I believe) and does not use load-sharing suspension systems like trucks.

Pulled directly from the ADR database.

ADR	Clause	Requirement	**Verification category** (for systems)
43	8.1	All the ‘Axles’ in an ‘Axle Group’ other than a ‘Twin Steer Axle Group’ must be related to each other through a ‘Load Sharing Suspension’ except:	Out of scope/Not applicable

43	8.1.1	This requirement must not apply to a 'Close Coupled Axle Group' on any motor vehicle or trailer (including a 'Semi-trailer') with a 'GVM' or 'ATM' less than 4.5 tonne provided that the load carrying capacity of each of the 'Axles' and the wheel and tyres fitted to each 'Axle' is at least 120% of the nominal load imposed on that 'Axle' with the vehicle or trailer at its 'GVM' or 'ATM';	Out of scope/Not applicable
43	8.1.2	This requirement must not apply to a 'Close Coupled Axle Group' fitted with a 'Retractable Axle' on any motor vehicle or trailer (including a 'Semi-trailer') with a 'GVM' or 'ATM' less than 4.5 tonne provided that the load carrying capacity of each other 'Axle' and the wheels and tyres fitted to it is at least equal to the load imposed on that 'Axle' with the motor vehicle or trailer at its 'GVM' or 'ATM' and with the 'Retractable Axle' retracted	Out of scope/Not applicable